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RESEARCH ARTICLE

FUSCANet: Enhancing Skin Disease Classification Through Feature Fusion and Spatial-Channel Attention Mechanisms

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ABSTRACT Skin diseases represent a prevalent global health issue that significantly impacts the physical and mental well-being of patients. With the widespread application of computer vision technology in dermatology, automating skin lesion classification through computer algorithms has become a crucial method for improving diagnostic efficiency and reducing the mortality rate due to malignant skin conditions. In response to this need, a novel lightweight neural network model, called Feature fUsion and Spatial-Channel Attention Network (FUSCANet) model, is proposed in this paper, based on the MobileViT framework, aiming at classifying multi-class skin disease images on mobile or embedded devices. Firstly, a newly designed Leaky MobileNetV2 (LMV2) block, integrated into the proposed model, allows to enhance its ability to extract features from skin images. Then, a novel Multi-Scale Feature Aggregation (MSFA) layer is added to enhance feature extraction across various scales, effectively capturing more comprehensive features and fusing feature maps from different scales to improve final feature reuse. Lastly, a newly designed Enhanced Convolutional Block Attention Module (ECBAM) increases the model's focus on important information, allowing for the final extraction and summarization of features. The proposed FUSCANet model is evaluated on four different datasets (the public PAD-UFES-20, HAM10000, and ISIC 2019 datasets, and a private dataset) containing images of various skin diseases. The obtained experimental results show that FUSCANet outperforms existing models on most evaluation metrics, while maintaining low parameter counts, making it suitable for deployment in resource-constrained environments.

INDEX TERMS Skin diseases, classification, MobileViT, feature fusion, spatial-channel attention.

I. INTRODUCTION

The skin, as the body's external covering and the organ directly exposed to the environment, is particularly vulnerable to external impacts, which can lead to various skin diseases. These diseases, affecting the epidermis, dermis, and subcutaneous tissue, are caused by a variety of factors including physical and chemical irritants, pathogenic infections, endocrine and metabolic disruptions, immune responses, and degenerative changes. Some skin diseases, beyond their physical manifestations, can also have significant psychological impacts on patients [1], [2]. Globally,

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skin diseases are the fourth leading cause of non-fatal disease burden [3].

The widespread use of computer vision technology in analyzing optical images of skin diseases has made the diagnosis of skin lesions faster and more accurate [4]. Skin diseases pose a substantial health threat worldwide, and automating the classification of skin lesions through computer algorithms is a crucial way to improve diagnostic efficiency and reduce mortality from malignant skin conditions [5]. Neural-network-based algorithms for skin disease classification have gained significant attention, particularly as aging populations and the prevalence of chronic diseases increase the demand for healthcare resources. These algorithms not only assist doctors in making quicker and more accurate diagnoses but also enhance clinical efficiency. Additionally, they contribute to improving the quality of medical services and hold promises for advancing intelligent healthcare, promoting telemedicine, and optimizing the use of healthcare resources.

Globally, large-scale skin disease image datasets have been established, providing a solid foundation for deep learning research. For instance, the 2020 Society for Imaging Informatics in Medicine (SIIM) − International Skin Imaging Collaboration (ISIC) Melanoma Classification challenge dataset [6] offers labeled data for training deep learning models. Another example is the PAD-UFES-20 dataset [7], which includes skin disease images captured by various smartphones. These datasets enable researchers to test and refine their models, fostering rapid advancements in neural-network-based intelligent diagnostic technologies.

In this paper, we detail the design and training methods utilized by the proposed model and demonstrate its effectiveness in skin disease image classification tasks through experiments. We aim to contribute new directions and solutions to the development of intelligent diagnostic support technologies. Based on an improved MobileViT [8], the proposed model introduces several innovations. Firstly, the original model's modules are enhanced to prevent neuron failure during feature extraction, thereby minimizing information loss and improving feature extraction efficiency of the model. Secondly, a novel feature fusion layer is designed to differentiate the importance of various features while integrating multilevel features, using shallow features to enrich deeper ones. Finally, a newly designed attention mechanism is incorporated into the model, enabling more accurate localization of lesion areas and better distinction between different types of skin diseases.

The conducted experiments have shown that the proposed FUSCANet model outperforms existing models on most evaluation metrics used, while maintaining low parameter counts, making it suitable for deployment in resource-constrained environments. FUSCANet has broad application prospects as it can not only improve clinical diagnostic efficiency but also can provide strong support for the early detection and intelligent treatment of skin diseases.

II. RELATED WORK

A. SKIN LESION IMAGE CLASSIFICATION

In the early stages of skin disease image classification technology [9], manual feature extraction techniques were primarily relied upon color and texture. These features were then input into traditional classifiers like decision trees and support vector machines (SVMs). These methods played a significant role in early skin lesion classification tasks. For example, Alencar et al. [10] proposed a system for classifying dermoscopic images on mobile devices. They utilized asymmetry, border, color, and diameter (the so-called ABCD rule) to analyze skin lesions, employing a multilayer perceptron network (MLP) trained with backpropagation for feature classification. This method improved the accuracy of dermoscopic image classification, particularly in mobile environments. Shrivastava et al. [11] proposed a computer-aided diagnosis (CAD) system specifically for psoriasis skin images. Their method involved principal component analysis (PCA) to extract relevant features, followed by classification using a SVM. This approach enhanced the accuracy of psoriasis diagnosis through a comparative study based on higher-order statistical, texture, and color features. Fekrache et al. [12] proposed a segmentation technique based on ant colony optimization, utilizing texture, relative color, and geometric properties as features of skin lesions. They then employed two traditional machine learning classifiers – artificial neural networks (ANNs) and k-nearest neighbors (KNN). The use of a dual classifier system improved the overall accuracy of melanoma and benign skin lesion classification. Despite the promising results achieved by these traditional machine learning methods, manually designing features requires extensive tuning to achieve optimal performance when handling overly complex datasets. With significant advancements in deep learning, CAD systems have made remarkable progress in skin disease classification and detection. Deep convolutional neural networks (CNNs) can automatically extract features from input data, enabling an end-to-end training. For example, Esteva et al. [13] demonstrated classification of skin lesions using a single CNN, trained end-to-end directly from images, using only pixels and disease labels as inputs. Hao et al. [14] leveraged image features extracted by a self-attention mechanism and fused the global patterns and long-range dependencies captured by a Swin Transformer with the local features extracted by a CNN, resulting in more discriminative feature representations and improved classification accuracy. By learning high-level abstract features, deep CNNs can extract information from large-scale image data, thereby achieving accurate classification of skin lesions.

B. APPLICATION OF VISION TRANSFORMER IN MEDICAL IMAGE PROCESSING

In recent years, with the significant advancements in deep learning technologies represented by CNNs, there have been substantial developments in the field of medical imaging.

CNNs and their variants have achieved state-of-the-art performance in several medical imaging-assisted diagnostic tasks. However, CNNs tend to overlook long-range dependencies in images, such as the non-local relationships between objects or lesion areas within images. To address this, Google proposed the Transformer model [15] based on self-attention mechanism modules, which has commonly been used to replace recurrent neural networks (RNNs) in natural language processing (NLP) tasks. Transformer can address sequential learning problems all at once and effectively stack very deep models. Inspired by the success of Transformers in NLP, the Vision Transformer (ViT) model [16] treats image classification as a sequence prediction task of image patches, thereby capturing long-range dependencies in the input images. This marks the first successful application of Transformer networks in the field of computer vision. The introduction of ViT has provided a novel approach to medical image recognition, demonstrating an outstanding performance. For example, Shome et al. [17] proposed a ViT-based architecture for chest X-ray image processing. To address the issue of insufficient data, their study merged three open-source COVID-19 datasets into a single dataset containing 30,000 images. Their network architecture demonstrated excellent performance in both binary and multi-class classification tasks. Vachmanus et al. [18] proposed a deep learning framework, based on ViT, for classifying skin lesions using both images and patient metadata. This study evaluated the classification performance of skin lesions across four public skin lesion image datasets and achieved outstanding results. Ayas et al. [19] proposed an end-to-end Swin Transformer model for multi-class skin lesion classification that does not require prior knowledge. Evaluated on the ISIC 2019 Skin Lesion Analysis Towards Melanoma Detection Challenge dataset, this model achieved the highest balanced accuracy among the models compared to performing multi-class skin lesion classification. Increasing evidence from ViT-based studies highlights the significant advantages of ViT in the field of image processing.

C. APPLICATION OF LIGHTWEIGHT MODELS IN MEDICAL IMAGE PROCESSING

Deep learning technologies primarily rely on CNNs, which require substantial hardware and software resources to build. As a result, the application scenarios for deep CNNs are limited. In many underdeveloped regions the use of limited medical equipment is common practice. Therefore, researching how to construct lightweight yet high-performance network models is particularly important for promoting their widespread use on low-end devices. The emergence of lightweight neural networks addresses the challenge of applying traditional deep learning to mobile devices with limited capabilities and has also demonstrated excellent performance in skin disease classification. For example, Hao et al. [20] proposed a lightweight segmentation model based on an encoder–decoder architecture, called GSCEU-Net, for skin lesion image segmentation tasks. The conducted experiments across multiple datasets indicated that the proposed model achieved excellent performance in skin lesion image segmentation while also demonstrating strong competitiveness in terms of lightweight performance. Ruan et al. [21] proposed the MALUNet model, which combines four modules with a U-shaped architecture to create a lightweight network structure for skin lesion segmentation tasks. Experiments conducted on two public datasets demonstrated that the model achieved state-of-the-art results in balancing parameter counts, computational complexity, and segmentation performance. In the field of image classification, lightweight models, such as MobileNet [22], ShuffleNet [23], and MobileViT [8], have achieved excellent classification performance while maintaining a small number of parameters. Srinivasu et al. [24] proposed a computerized process for skin disease classification using a combination of deep learning-based MobileNetV2 and Long Short-Term Memory (LSTM) networks. Experiments demonstrated that their approach allows to reduce computational complexity while providing greater robustness in identifying affected regions. Ji et al. [25] proposed LightCF-Net, a lightweight network for real-time polyp segmentation, which integrates a large kernel attention with a channel attention in the encoder, a visual attention Mamba module in skip connections, and a pyramid split attention module in the bottleneck. It effectively captures long-range dependencies while maintaining efficiency and achieves good segmentation performance with low computational cost on four public datasets.

D. APPLICATION OF FEATURE FUSION IN MEDICAL IMAGE PROCESSING

Feature fusion has become an essential technique in medical image processing, particularly in object detection tasks. By combining features from different layers or modalities, fusion methods enhance the model's ability to detect lesions, tumors, and other clinically relevant targets with greater accuracy and robustness. In medical object detection, effective fusion of low-level texture details and high-level semantic information helps localize small or ambiguous targets, which is critical for early diagnosis and treatment planning. Hao et al. [26] proposed YOLO-CXR, a detection network tailored for locating multiple small lesions in chest X-ray images, with a strong emphasis on effective feature fusion. The network incorporates a Selective Feature Fusion strategy that adaptively combines features at different scales, significantly enhancing the detection of small and complex lesions. Additionally, RefConv layers are employed to improve multi-disease feature representation, while the Efficient Channel and Local Attention mechanism enhances spatial sensitivity. Experimental results demonstrate that YOLO-CXR outperforms existing networks, particularly in detecting small lesions. Liu et al. [27] proposed SOCR-YOLO, a novel detection model for small object detection in medical images. The model first employs a Bi-FPN structure

to enhance multi-scale feature fusion, improving its ability to detect lesions of different sizes. It also introduces the SOCR module, which combines shuffled channel operations with online convolutional re-parameterization to enrich feature representation while maintaining efficiency. In addition, the loss function is refined to optimize the training process. Experimental results demonstrate that SOCR-YOLO achieves higher accuracy and lower computational cost compared to several advanced detection models, particularly in detecting small and subtle lesions. In summary, most feature fusion strategies applied to medical image processing are task-specific, aiming to effectively combine features at different levels to improve representation and performance.

III. PROPOSED MODEL

A. OVERALL STRUCTURE

A lightweight skin disease classification model, named FUSCANet, is proposed here based on MobileViT. FUSCANet integrates multi-feature fusion and attention mechanisms, as illustrated in Figure 1. The model consists of five feature extraction (FE) layers and a Multi-Scale Feature Aggregation (MSFA) layer. The FE layers are composed of the original MobileViT blocks, the Inverted Residual blocks of MobileNetV2 (denoted as MV2 in Figure 1), and an improved version of the Inverted Residual block (denoted as LMV2 in Figure 1). The newly designed MSFA layer reshapes and integrates multiscale feature attention weights, enhancing the dependency between local and global features and enriching the feature information. The model's final layers have been improved with the introduction of a novel Enhanced Convolutional Block Attention Module (ECBAM), which helps avoid the adverse effects of unnecessary channel compression. Finally, the feature information is normalized and used for prediction through a linear layer.

B. LMV2 BLOCK

The MobileViT feature extraction layer includes multiple MobileNetV2 (MV2) blocks with a design structure that incorporates a depthwise separable convolution, inverted residuals, and a linear bottleneck. This structure allows to effectively extract local features while also reducing the number of model parameters.

Unlike the residual structure in ResNet models, which first reduces and then increases the feature dimensions, the MV2 block employs an inverted residual structure. Initially, the block uses a 1×1 pointwise convolution to expand the feature dimensions, thus enhancing feature granularity. This is followed by a 3×3 depthwise convolution to extract features from each channel, thereby reducing the number of parameters. The final step involves another 1×1 pointwise convolution to reduce the feature dimensions, decreasing the model's output dimensionality. To prevent the loss of low-dimensional feature information that might occur due to the activation function, a linear activation function is applied after the final 1×1 pointwise convolution. Residual connections in the MV2 block are only applied under specific conditions, i.e., when the stride is 1, and the input and output dimensions are the same. Introducing residual connections in these scenarios helps prevent feature loss. When the stride is 2, a concatenation connection is used to achieve feature downsampling.

In the original MV2 block, the activation function used is the ReLU activation function, which is defined as follows:

$$f(x) = max(0, x) \tag{1}$$

where $f(x)$ denotes the output of the ReLU function and x denotes the input value to the function.

However, to accommodate the lower precision of mobile devices and enhance the numerical resolution, the maximum

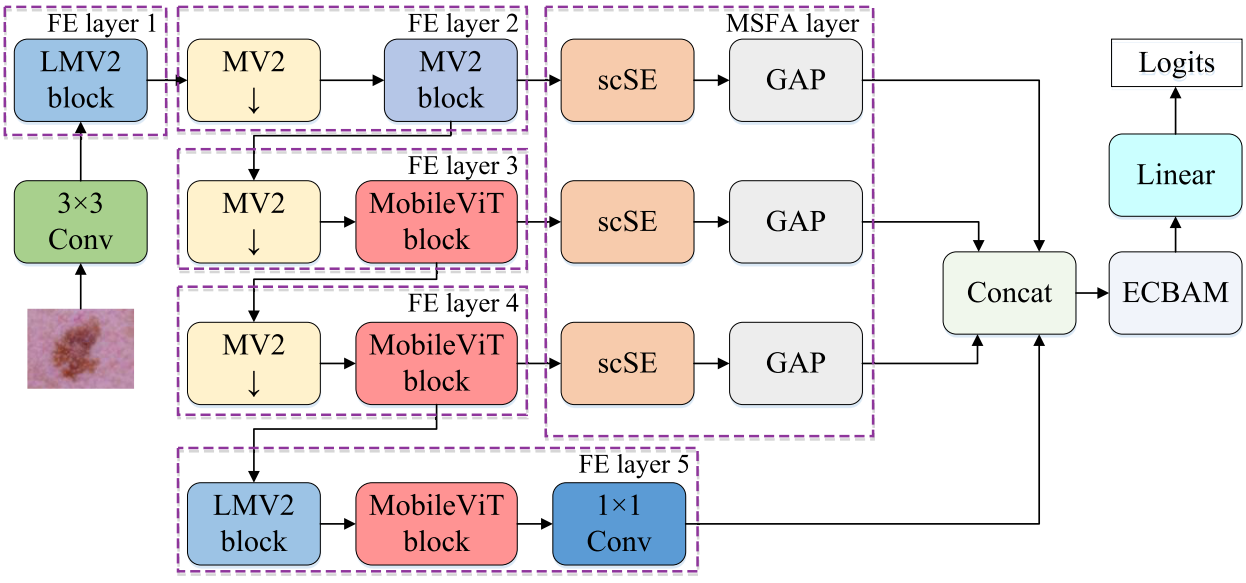


FIGURE 1. The proposed FUSCANet model's structure.

output of the ReLU activation function was capped at 6. While this adjustment mitigates precision loss on low-precision devices, the ReLU activation function can lead to issues such as neuron death. When the input value is negative, the gradient of the ReLU function becomes zero, which hinders the update of weights. If neuron death occurs during feature extraction in the shallow network layers, it can result in information loss. This is particularly concerning when extracting low-level features such as color and texture from skin disease images, as these features are primarily captured by the shallow network layers. Therefore, preventing neuron death in these layers is crucial. In contrast, the deep network layers, where the MV2 blocks reside, are responsible for extracting more complex and abstract features. If information loss occurs in these layers, it can severely affect the model's ability to extract and recognize these critical features. To avoid these negative impacts, we replaced the ReLU activation function following the 3×3 depthwise convolution in the MV2 block with the Leaky ReLU activation function, defined as follows:

$$f(x) = max(\alpha x, x) \tag{2}$$

where α is a small positive constant (typically set to 0.01), allowing a small, non-zero gradient when the input is negative.

The Leaky ReLU activation function is a variation of the ReLU activation function, designed specifically to address the issue of the ReLU function outputting zero for negative input values. The introduction of α allows the function to have a small negative slope in the negative input region, rather than being set to zero, as illustrated in Figure 2. Compared to the ReLU function, the Leaky ReLU function more readily activates neurons, thereby reducing the risk of neuron death.

After replacing the activation function, a newly designed LMV2 block was obtained, which was used in the FE layer 1 and FE layer 5 of the proposed model (c.f., Figure 1) to improve classification performance while keeping the same number of model parameters.

C. MSFA LAYER

In dermoscopic images, the lesion areas are often accompanied by noise elements such as shadows, bubbles, scale markers, and body hair. These noise factors lead to a dual negative effect: (1) high-frequency interference patterns (e.g., gradient discontinuities at body hair edges) are formed in the shallow feature space; and (2) spurious correlations in feature activations (e.g., incorrect associations between scale markers and lesion textures) occur in the deep semantic space. To effectively address this issue, an architectural innovation based on the MobileViT model is introduced here, proposing a solution that enhances the model's ability to focus on the key features of lesion areas.

In CNNs, the feature extraction capabilities vary across different layers. Shallow network layers primarily capture simple low-level features such as position and texture, which are crucial for detecting lesion locations. Deep network layers, with larger receptive fields and stronger semantic expression capabilities, extract more complex high-level features that are essential for precise skin lesion classification. To further enhance the model's ability to capture various features and effectively combat noise interference, we designed a novel feature fusion layer, named MSFA.

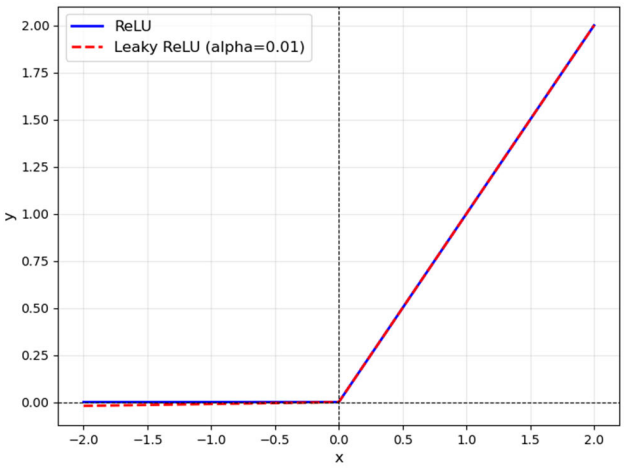


FIGURE 2. ReLU vs. Leaky ReLU.

This feature fusion layer combines outputs from multiple network layers, using three branches for feature fusion. Each branch includes a spatial and channel SE (scSE) attention module [28] and a global average pooling (GAP) layer. Specifically, the three branches receive outputs from FE layers 2, 3, and 4, respectively, and perform weighted feature fusion to emphasize key features in the lesion area. Through this approach, the MSFA layer not only refines features from different layers but also enhances the model's ability to resist noise interference, thereby improving its performance in skin lesion classification.

The scSE attention module, depicted in Figure 3, is an improved version of the SE attention module [29]. Its primary components are the Spatial Squeeze and Channel Excitation (cSE) block and the Channel Squeeze and Spatial Excitation (sSE) block. The cSE block introduces a channel attention mechanism that effectively integrates and enhances feature information along the channel dimension, similarly to traditional channel attention mechanisms like SE. The key difference is that the cSE block performs a dimension reduction followed by a dimension expansion on the obtained attention weights. This process effectively integrates channel information, simplifies the module's complexity, reduces computational load, and increases processing speed. The other block (sSE) focuses on enhancing and integrating information along the spatial dimension of the feature map. Similarly to the channel dimension approach, it first extracts weight information and then multiplies the weights with the original feature map to achieve attention enhancement. However, unlike the cSE block which uses GAP, the sSE block extracts weight information by unfolding the spatial

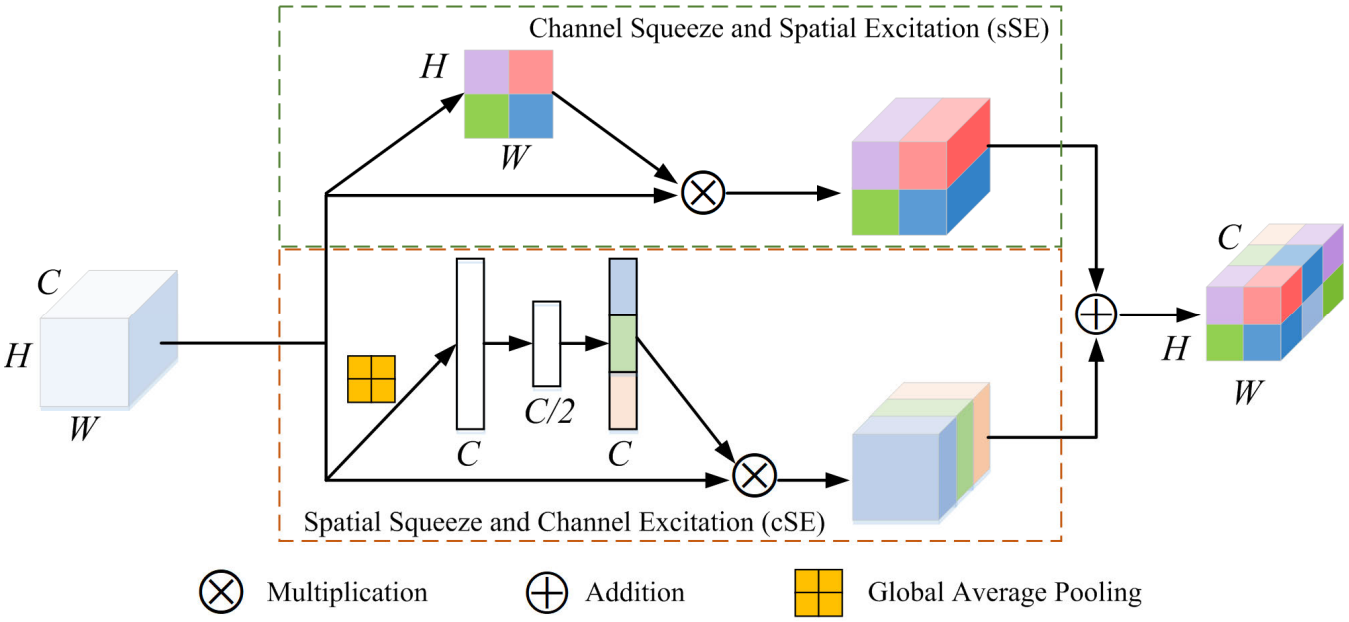


FIGURE 3. The scSE attention module, utilized by the proposed FUSCANet model.

dimension. It uses a convolution layer with an output channel of 1 and a kernel size of 1×1 for information integration, rather than relying on GAP.

The combination of the sSE block and the cSE block allows the scSE attention module to enhance and integrate information across both the spatial and channel dimensions. The feature results from both blocks are then summed along the channel dimension.

The sSE processing can be expressed as follows:

$$\boldsymbol{Y}_{\boldsymbol{sSE}} = [\sigma\left(\boldsymbol{q}_{1,1}\right)\boldsymbol{x}^{1,1}, \cdots, \sigma\left(\boldsymbol{q}_{i,j}\right)\boldsymbol{x}^{i,j}, \cdots, \sigma\left(\boldsymbol{q}_{H,W}\right)\boldsymbol{x}^{W,H}] \tag{3}$$

where $\boldsymbol{x}^{i,j} \in \mathbb{R}^{1\times1\times C}$ denotes the slice of the input tensor $\boldsymbol{X} = [\boldsymbol{x}^{1,1}, \boldsymbol{x}^{1,2}, \cdots, \boldsymbol{x}^{i,j}, \cdots, \boldsymbol{x}^{W,H}]$ corresponding to the spatial position $(\boldsymbol{i},\boldsymbol{j})$, where $\boldsymbol{i} \in \{\boldsymbol{1}, \boldsymbol{2}, \cdots, \boldsymbol{W}\}$ and $\boldsymbol{j} \in \{\boldsymbol{1}, \boldsymbol{2}, \cdots, \boldsymbol{H}\}$, $\boldsymbol{H}$, $\boldsymbol{W}$, and $\boldsymbol{C}$ denote the height, width, and number of channels of the input tensor $\boldsymbol{X}$, respectively, $\sigma(\cdot)$ denotes the sigmoid function, and $\boldsymbol{q}_{i,j}$ denotes a single element at the spatial position $(\boldsymbol{i},\boldsymbol{j})$ in the spatially compressed tensor $\boldsymbol{q} \in \mathbb{R}^{H\times W}$, obtained as follows:

$$\boldsymbol{q} = \boldsymbol{W}_{\boldsymbol{sq}} * \boldsymbol{X} \tag{4}$$

where $\boldsymbol{W}_{\boldsymbol{sq}}$ denotes a convolutional kernel with dimensions $\boldsymbol{1}\times\boldsymbol{1}\times\boldsymbol{C}\times\boldsymbol{1}$ and symbol $*$ denotes the convolution operation.

The sSE module divides the input along the channel dimension, emphasizing the overall importance of each channel. In contrast, the cSE module splits the input according to spatial locations, treating the input vector $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_C]$ as a combination of channels $\boldsymbol{x}_i \in \mathbb{R}^{H\times W}$. To achieve spatial compression, GAP is applied to the split input vector $\boldsymbol{X}$, generating $\boldsymbol{X'} = [\boldsymbol{x'}_1, \boldsymbol{x'}_2, \cdots, \boldsymbol{x'}_C]$, which contains global spatial information. The scalar value $\boldsymbol{x'}_k$, generated by applying GAP to the $\boldsymbol{k}$-th channel, representing the global average feature of that channel, is obtained as follows:

$$\boldsymbol{x'}_k = \frac{1}{\boldsymbol{H}\times\boldsymbol{W}} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{x}_k \tag{5}$$

Next, the dependencies between channels are compiled into $\hat{\boldsymbol{X}} = [\hat{\boldsymbol{x}}_1, \hat{\boldsymbol{x}}_2, \cdots, \hat{\boldsymbol{x}}_C]$, as follows:

$$\hat{\boldsymbol{X}} = \boldsymbol{W}_1(\delta(\boldsymbol{W}_2\boldsymbol{X'})) \tag{6}$$

where $\boldsymbol{W}_1 \in \mathbb{R}^{C\times\frac{C}{2}}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{\frac{C}{2}\times C}$ denote the weight matrices, and $\delta(\cdot)$ denotes the ReLU activation function. Next, the dependencies between channels are compiled into $\hat{\boldsymbol{X}}$ as per (7) and the final output of the cSE module is obtained as per (8):

$$\boldsymbol{Y}_{\boldsymbol{cSE}} = [\sigma\left(\hat{\boldsymbol{x}}_1\right)\boldsymbol{x}_1, \sigma\left(\hat{\boldsymbol{x}}_2\right)\boldsymbol{x}_2, \cdots, \sigma\left(\hat{\boldsymbol{x}}_C\right)\boldsymbol{x}_C] \tag{7}$$

$$\boldsymbol{Y} = \sum_{k=1}^{C} \frac{1}{\boldsymbol{H}_k\times\boldsymbol{W}_k} \sum_{i=1}^{H_k} \sum_{j=1}^{W_k} (\boldsymbol{Y}_{\boldsymbol{sSE_k}} + \boldsymbol{Y}_{\boldsymbol{cSE_k}}) \tag{8}$$

The MSFA layer compensates for the limitations of baseline's feature expression at different layers, enhancing the model's global modeling capabilities while retaining detailed features. Through information interaction and feature enhancement, MSFA improves the representation of lesion regions, providing more discriminative features for skin disease classification. However, this strategy also leads to a slight increase in the model's parameter counts and floating-point operations per second (FLOPS), though this cost is justified by the performance improvements.

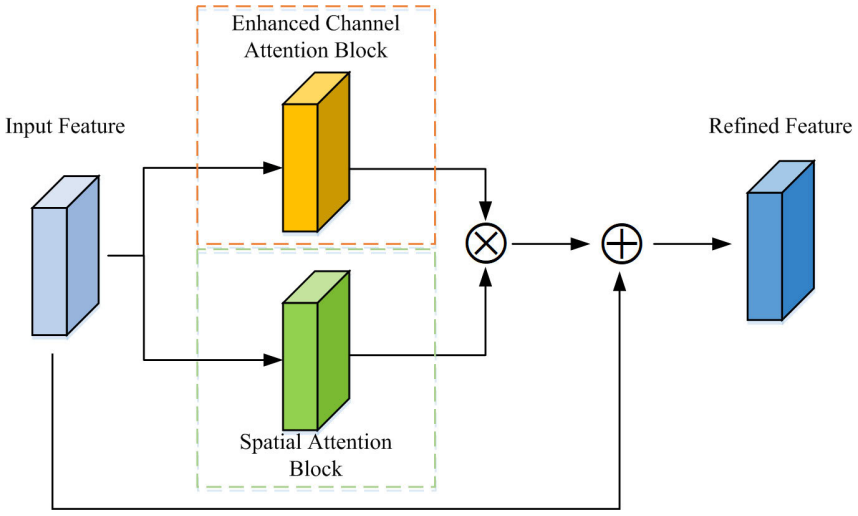


FIGURE 4. The novel ECBAM structure, utilized by the proposed FUSCANet model.

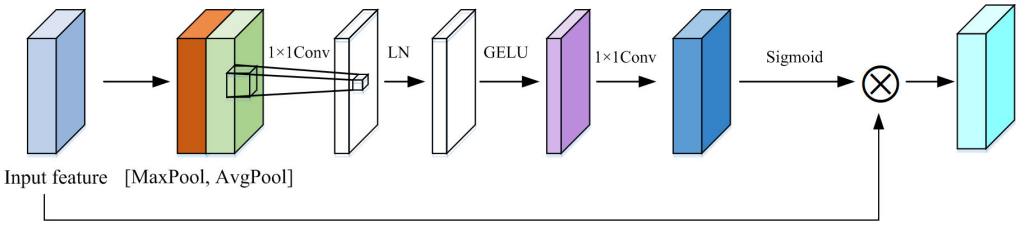


FIGURE 5. The enhanced CAM, utilized by the novel ECBAM.

D. ECBAM

The Convolutional Block Attention Module (CBAM) enhances the perceptual capability of CNNs by introducing channel and spatial attention, thereby improving performance without increasing network complexity. Inspired by this, we propose an enhanced CBAM structure shown in Figure 4, resulting in a novel module called ECBAM. Compared to CBAM, ECBAM utilizes an enhanced Channel Attention Module (CAM), illustrated in Figure 5.

In skin disease images, the feature distribution of different lesion types is often complex and diverse, so effectively extracting and enhancing deep feature information is crucial. To this end, the enhanced CAM retains the parallel methods of maximum pooling and average pooling, and by concatenating the outputs of the two pooling methods, maximizes the preservation of the information in the original feature map, thereby enhancing the model's ability to capture key lesion areas. In addition, the enhanced CAM replaces the fully connected layers in the original CAM with 1×1 convolutional layers, reducing the number of parameters, and introduces layer normalization (LN) to balance the feature distribution of batch data.

Moreover, to prevent neuron death and enhance the model's ability to learn complex features, the Gaussian Error Linear Unit (GELU) function is used as an activation function in the enhanced CAM. The output of the GELU activation function approximates a Gaussian distribution when the input is near zero, which helps improve the generalization ability of the neural network, making the model more adaptable to different data distributions. The overall processing of the enhanced CAM can be expressed as follows:

$$F_C = \sigma(W_1(GELU(LN(W_0(Concat(AvgPool(X), MaxPool(X)))))) \otimes X \quad (9)$$

where $LN(\cdot)$ denotes a layer normalization, W_0 and W_1 denote the two weight matrices, $Concat(\cdot)$ denotes a concatenation along the channel dimension, $AvgPool(\cdot)$ denotes a global average pooling, $MaxPool(\cdot)$ denotes a max pooling, and symbol $\otimes$ denotes an element-wise multiplication.

In addition to the enhanced CAM, ECBAM also utilizes a Spatial Attention Module (SAM) with the same original structure as in CBAM. SAM focuses on identifying which features are important in the spatial dimension, complementing the channel attention. The SAM processing can be represented as follows:

$$F_S = \sigma(f^{7\times7}(Concat(AvgPool(X), MaxPool(X)))) \quad (10)$$

where $f^{7\times7}(\cdot)$ denotes a convolution operation with a kernel size of 7×7.

ECBAM was incorporated into the proposed model to perform hybrid attention, combining the channel attention with spatial attention. Placing ECBAM before the linear layer of the proposed model allows it to enhance feature representation and capture critical information in skin disease images before mapping features to the class space. This process also helps the model suppress irrelevant or redundant features, thereby improving its classification performance. The overall ECBAM processing can be expressed as follows:

$$\boldsymbol{Y} = \boldsymbol{X} \oplus \boldsymbol{F_S} \otimes \boldsymbol{F_C} \tag{11}$$

where $\otimes$ denotes an element-wise multiplication and $\oplus$ denotes an element-wise addition.

IV. EXPERIMENTS AND RESULTS

A. DATASETS

In the conducted experiments, four datasets were used to evaluate the performance of the proposed FUSCANet model in comparison to other models, both baseline and advanced ones. These four datasets each have their own characteristics, which allows to evaluate the model performance more convincingly.

The first dataset was PAD-UFES-20, which contains 2,298 skin disease images from 1,373 patients, captured by various smartphones, belonging to six classes: three classes of skin cancer – basal cell carcinoma (BCC), squamous cell carcinoma (SCC), and melanoma (MEL), and three classes of skin diseases – actinic keratosis (ACK), seborrheic keratosis (SEK), and nevus (NEV). The SCC class also includes Bowen's disease (BOD). Due to the fact that the images in this dataset are captured by smartphones, their resolution, lighting conditions, and background noise have significant diversity, which can be used to effectively test the robustness of models in real scenarios and their adaptability to low-quality images.

The second dataset was HAM10000 [30] – a large-scale dataset, containing multi-source dermatoscopic images of common pigmented skin lesions, provided for the ISIC 2019 challenge. This dataset contains 10,015 skin disease images of the following classes: actinic keratosis and intraepithelial carcinoma (AKIEC), BCC, benign keratosis-like lesions (BKL), dermatofibroma (DF), MEL, melanocytic nevus (NV), and vascular skin lesions (VASC). The large number of images and significant class imbalance make this dataset suitable for testing the generalization ability of models on large-scale data and their ability to handle class imbalance issues. In addition, dermatoscopy images have high resolution and clear lesion features, which allows to verify the model classification accuracy and feature extraction ability on high-quality medical images.

The third dataset was a private dataset, provided by Peking Union Medical College Hospital. It includes 2,900 anonymized images of six classes of skin lesions, as follows: 1,615 images of acne (AC), 125 images of discoid lupus erythematosus (DLE), 400 images of melasma (ME), 280 images of Ota nevus (ON), 310 images of rosacea (ROS), and 170 images of seborrheic dermatitis (SD). Sample images of the various skin lesions from this dataset are shown in Figure 6. The main characteristic of this dataset is the high inter-class similarity, which allows to effectively test the model fine-grained classification ability in distinguishing similar classes. In addition, the images in the private dataset are sourced from actual clinical scenarios, which allows to verify the model applicability and robustness in real-life medical environments.

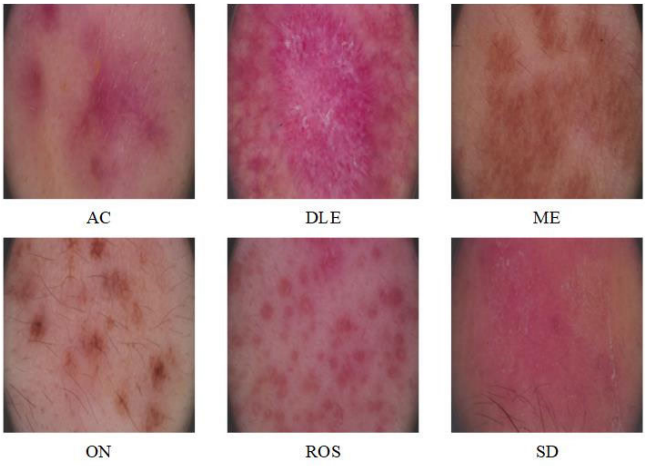


FIGURE 6. Sample images of the private dataset.

The fourth dataset, used in the experiments, was the large-scale ISIC 2019 dataset (https://www.kaggle.com/datasets/andrewmvd/isic-2019) containing the training data for the ISIC 2019 challenge, including data from previous challenges in 2018 and 2017. It includes 25,331 labeled images, classified into eight classes: actinic keratosis (AK), BCC, BKL, DF, ME, NV, SCC, and VASC.

We applied dataset preprocessing and data augmentation to the images of all four datasets. First, to ensure consistent input size and help a model to focus on key regions, the original images were resized to 224 $\times$ 224 pixels. Then, data augmentation techniques such as random rotation and histogram equalization were performed on the training set to increase the diversity of the image information and mitigate overfitting during the model training.

However, the use of the PAD-UFES-20 dataset faces challenges due to its small size and significant class imbalance. Previous research [31] indicates that such an imbalance can significantly reduce the training performance. Although the average evaluation metrics across all classes may be high, class imbalance can lead to lower evaluation results for classes with fewer images. To address this issue in the PAD-UFES-20 dataset, which consists of images captured by smartphones, oversampling methods (e.g., horizontal flipping, random cropping, Gaussian blur, linear contrast, random translation, rotation, and minor shearing) were employed to balance the data for each class and generate new images from the existing ones. After performing this data augmentation, the number of images in the PAD-UFES-20 dataset was

increased by 2.5 times. The number of images in each class before and after performing the data augmentation is shown in Table 1.

Due to class imbalance, extensive data augmentation was applied to the PAD-UFES-20 dataset. While data augmentation helps balance the dataset, it can introduce biases, such as overfitting to augmented patterns that may not fully represent real-world variations. An alternative, Generative Adversarial Networks (GANs), could generate more realistic samples, potentially offering better data diversity. However, GANs are computationally expensive and require complex training, making them less practical for large-scale real-world applications. In contrast, data augmentation is simple, efficient, and preserves the natural distribution of data, making it a more suitable approach for improving the model performance in practical settings.

To ensure fairness and rigor in the experiments, each dataset was randomly split into a training set and a testing set in an 8:2 ratio.

TABLE 1. Number of PAD-UFES-20 images in each class before and after performing data augmentation.

Data augmentation	BCC	SCC	MEL	ACK	SEK	NEV	**Total**
Before	845	192	52	730	235	244	**2298**
After	958	958	958	958	958	958	**5748**

B. EXPERIMENTAL SETUP

We employed the PyTorch deep learning framework to construct the proposed FUSCANet model. The programming language used was Python 3.8, operating on a Linux system. The hardware setup utilized an NVIDIA GeForce RTX 3090 GPU. During model training, all parameters for each experiment were strictly controlled to ensure consistency. All models had identical hyperparameter initialization and were optimized using the AdamW optimizer [32]. The cross-entropy loss was used as a loss function in all experiments, and the learning rate was adjusted using a cosine annealing strategy. The batch size for all experiments was set to 64, with an initial learning rate of 5e-4 and a minimum learning rate of 5e-10. The compared models were trained for 35 epochs. Additionally, all models were pre-trained on ImageNet [33] and initialized with pre-trained parameters.

C. EVALUATION METRICS

To evaluate the classification performance of the compared models, we used accuracy, precision, recall, specificity, and F1 score, calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

$$Specificity = \frac{TN}{TN + FP} \tag{15}$$

$$F1_{score} = \frac{2 \times precision \times recall}{precision + recall} \tag{16}$$

where ***TP*** denotes the true positive count, ***TN*** denotes the true negative count, ***FP*** denotes the false positive count, and ***FN*** denotes the false negative count.

In addition, we evaluated the computational complexity of models and their practicality of usage, based on the number of their parameters and FLOPS.

D. ABLATION STUDY EXPERIMENTS

To validate the effectiveness of each newly designed component, utilized by the proposed FUSCANet model, we conducted ablation study experiments on the PAD-UFES-20 and HAM10000 datasets by using MobileViT as a baseline. Tables 2 and 3 detail the combinations of various components across different model versions (with the best results highlighted in **bold**).

Three sets of ablation study experiments were conducted on each dataset. In the first set, the impact of adding each newly designed component (ECBAM, LMV2, MSFA) to the baseline was evaluated independently from the other components.

This set of experiments demonstrated that each of these components contributed positively to enhancing the baseline's classification performance, with MSFA and LMV2 making the most significant contribution on PAD-UFES-20 and HAM10000, respectively. Notably, the newly designed LMV2 block improved the classification performance of the baseline without increasing its parameter counts. On the PAD-UFES-20 dataset, adding MSFA to the baseline resulted in improvements across all five evaluation metrics, with the most notable gains in overall accuracy and average F1 score, which increased by 0.70 and 0.76 percentage points, respectively, compared to the baseline. Importantly, the model's parameter counts only increased by 0.01M, highlighting the efficiency of MSFA in enhancing performance with minimal computational cost.

In the second set of experiments, the newly designed components were added in pairs to the baseline. The combined use of MSFA and ECBAM produced the best results on PAD-UFES-20, with a small increase in FLOPS as well. This indicates that the combination of attention mechanisms and feature fusion enables the model to capture more effectively the correlations between features. On HAM10000, the best performing double combination was 'MSFA + LMV2', according to three out of five evaluation metrics used.

In the final experiment, all three newly designed components were added together to the baseline, resulting in the proposed FUSCANet model. This led to improvements across all metrics on PAD-UFES-20, while keeping the same number of model parameters and FLOPS compared to the best performing double combination ('MSFA + ECBAM'). On HAM10000, in the final experiment, improvements were recorded on three out of five evaluation metrics. This suggests

TABLE 2. Ablation study results, obtained on the PAD-UFES-20 dataset.

Model versions	Evaluation metrics (%)					Parameter counts (M)	FLOPS (M)
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score		
MobileViT (baseline)	92.58	92.59	92.61	98.52	92.44	**4.94**	1464.38
Baseline + ECBAM	93.11	93.11	93.04	98.62	93.01	5.09	1464.56
Baseline + LMV2	93.19	93.19	93.10	98.64	93.11	**4.94**	1464.38
Baseline + MSFA	93.28	93.28	93.24	98.66	93.20	4.95	1465.30
Baseline + ECBAM + LMV2	93.54	93.54	93.47	98.71	93.48	5.10	1464.56
Baseline + MSFA + LMV2	93.63	93.63	93.58	98.73	93.58	4.95	1465.31
Baseline + MSFA + ECBAM	93.98	93.98	93.98	98.80	93.93	5.27	**1465.69**
Baseline + MSFA + ECBAM + LMV2 (i.e., **FUSCANet *proposed***)	**94.59**	**94.59**	**94.66**	**98.92**	**94.56**	5.27	**1465.69**

TABLE 3. Ablation study results, obtained on the HAM10000 dataset.

Model versions	Evaluation metrics (%)				
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score
MobileViT (baseline)	91.15	87.32	85.63	97.76	86.39
Baseline + ECBAM	91.40	82.51	87.59	97.67	84.70
Baseline + LMV2	91.70	**94.91**	86.72	97.91	85.75
Baseline + MSFA	91.70	84.52	86.59	97.89	85.51
Baseline + ECBAM + LMV2	92.05	85.85	87.45	97.79	86.36
Baseline + MSFA + LMV2	92.20	88.81	86.64	**98.01**	87.50
Baseline + MSFA + ECBAM	92.25	84.89	86.62	97.84	85.62
Baseline + MSFA + ECBAM + LMV2 (i.e., **FUSCANet *proposed***)	**92.80**	87.41	**88.43**	97.99	**87.78**

that integrating all three newly designed components with the baseline helps the model learn richer feature information from skin lesion images.

In order to verify the improvement effect of introducing different attention mechanisms into the newly designed MSFA layer on the model performance in classifying skin disease images, we selected four commonly used attention mechanisms, namely the SE, ECA, CA, and scSE, and conducted experiments to determine the most suitable among these for multi-scale feature fusion. The corresponding classification performance, parameter counts, and inference time, achieved by the baseline with an added MSFA layer (denoted as 'Baseline + MSFA' in Tables 2 and 3) on the ISIC 2019 dataset, is shown in Table 4. The inference time was obtained on GTX 1050 Ti GPU (4GB). The obtained experimental results show that employing the scSE attention mechanism allows to achieve the highest overall performance in classifying skin disease images with minor detriments in FLOPS, parameter counts, and inference time, but reaching a good trade-off between performance gains and computational costs.

E. COMPARISON WITH BASELINE MODELS

To compare the classification performance of the proposed FUSCANet model with baseline models, experiments were conducted on four datasets – three public ones and a private one. The corresponding experimental results obtained are presented and discussed in the following subsections.

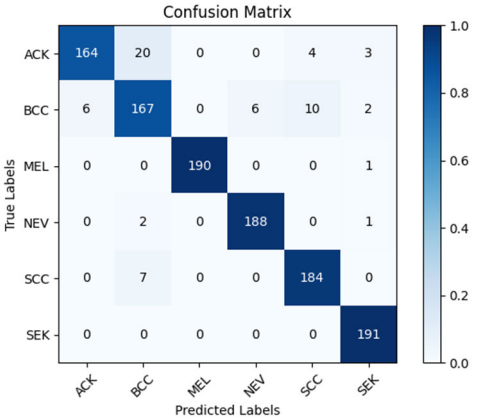


FIGURE 7. The confusion matrix of the proposed FUSCANet model on the PAD-UFES-20 dataset.

Figure 7 presents the FUSCANet classification performance on the PAD-UFES-20 public dataset in the form of a confusion matrix [34]. In this matrix, the accuracy of predicting each skin lesion class is represented by the diagonal values, while the intensity of the color of each square

TABLE 4. Classification performance on the ISIC 2019 dataset and computational complexity of the baseline (MobileViT) with added MSFA layer using different attention mechanisms.

Attention mechanisms	Evaluation metrics (%)					FLOPS (M)	Parameter counts (M)	Inference time (ms)
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score			
(BASELINE + MSFA) + SE	90.22	86.37	**83.94**	98.24	**85.05**	1465.00	4.95	9.37
(BASELINE + MSFA) + ECA	90.21	89.28	79.47	98.24	83.21	1464.99	**4.94**	**9.17**
(BASELINE + MSFA) + CA	90.26	85.08	81.84	98.29	83.02	**1465.68**	4.95	10.61
(BASELINE + MSFA) + SCSE (***CHOSEN***)	**90.44**	**90.06**	79.97	**98.30**	83.78	1465.30	4.95	9.64

TABLE 5. Classification performance and computational complexity comparison with baseline models on the PAD-UFES-20 dataset.

Models	Evaluation metrics (%)					Parameter counts (M)	FLOPS (M)
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score		
SHUFFLENET V2	91.60	88.24	86.01	97.91	87.05	**1.26**	151.69
MOBILENETV2	92.41	92.41	92.50	98.48	92.35	2.23	326.28
EFFICIENTNET B0	92.15	92.15	92.12	98.43	92.07	4.02	411.56
MOBILEVIT	92.58	92.59	92.61	98.52	92.44	4.94	1464.38
CONVNEXT TINY	93.89	93.89	93.90	98.78	93.89	27.80	**4454.77**
FUSCANET (*PROPOSED*)	**94.59**	**94.59**	**94.66**	**98.92**	**94.56**	5.27	1465.69

TABLE 6. Classification performance comparison with baseline models on the HAM10000 dataset.

Models	Evaluation metrics (%)				
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score
SHUFFLENET V2	90.80	86.49	82.67	97.79	84.13
MOBILENETV2	91.00	85.58	84.44	97.35	84.24
EFFICIENTNET B0	91.60	84.71	87.28	97.62	85.91
MOBILEVIT	91.15	87.32	85.63	97.76	86.39
CONVNEXT TINY	92.35	**88.45**	87.64	97.95	**87.93**
FUSCANET (*PROPOSED*)	**92.80**	87.41	**88.43**	**97.99**	87.78

indicates the accuracy for that class. As can be observed in Figure 7, the proposed FUSCANet performs particularly well in identifying the SEK and MEL classes. Additionally, FUSCANet achieved an overall accuracy of 94.59% across all classes, indicating an extraordinarily strong overall classification performance.

Table 5 presents a detailed comparison of the classification performance results of FUSCANet and baseline models, obtained experimentally on the PAD-UFES-20 public dataset. It can be observed that, aside from the parameter counts, FUSCANet outperforms all other models across all five evaluation metrics and FLOPS (except for ConvNeXt Tiny in terms of FLOPS). This indicates that the proposed model can classify skin lesions more accurately and faster. While ShuffleNet V2 has the smallest parameter counts, its performance across all evaluation metrics and FLOPS are relatively poor. Although FUSCANet has slightly higher parameter counts compared to other lightweight models, its value (5.27M) is only about 6.7% higher than that of MobileViT (4.94M), on which it was built with a minimal impact on the model complexity.

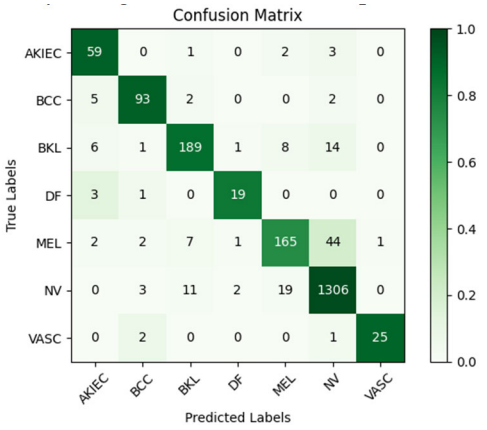


FIGURE 8. The confusion matrix of the proposed FUSCANet model on the HAM10000 dataset.

Moreover, the parameter counts of FUSCANet is several times lower than that of ConvNeXt Tiny. Therefore, the use of FUSCANet on mobile devices could have a significant positive effect for practical applications.

The HAM10000 dataset includes a greater variety of skin disease classes and a larger number of images, making it suitable for further evaluation of the model's performance and generalization ability. Figure 8 presents the classification performance of the proposed model on the HAM10000 public dataset in the form of a confusion matrix. As can be seen from that figure, the proposed FUSCANet also demonstrates excellent classification performance here, achieving an overall accuracy of 92.80% across all classes.

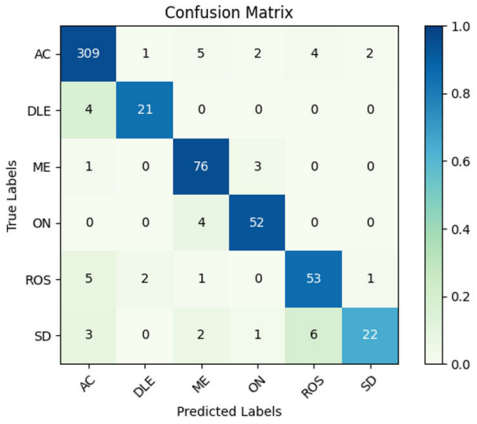


FIGURE 9. The confusion matrix of the proposed FUSCANet model on the private dataset.

Table 6 presents detailed comparison results of the classification performance of FUSCANet and the baseline models, obtained on the HAM10000 public dataset. It can be observed that FUSCANet is the winner, according to three out of five evaluation metrics used, while also holding second place on the other two metrics. While average recall and average F1 score of FUSCANet are slightly lower than those of ConvNeXt Tiny, it is important to note that the parameter counts of ConvNeXt Tiny is 5.28 times higher than that of FUSCANet. More importantly, FUSCANet demonstrates higher overall accuracy, compared to ConvNeXt Tiny. To evaluate the trade-off between model complexity and performance, we tested the inference time of both models on a GTX 1050 Ti GPU (4GB). The results show that FUSCANet achieves an inference time of 10.02685 ms, whereas ConvNeXt Tiny requires 14.776725 ms. This indicates that FUSCANet not only reduces computational resource consumption but also achieves a higher inference speed. The reduced latency is particularly advantageous for real-time applications, such as mobile-based skin disease diagnosis, where efficiency and responsiveness are crucial. Thus, FUSCANet provides a more balanced trade-off between performance and complexity, making it a more practical choice for deployment on resource-limited devices.

To further validate the generalization ability of FUSCANet, we tested its performance on a private dataset that features more homogeneous skin disease classes with smaller inter-class differences, primarily focusing on acne-like skin diseases. Figure 9 presents the classification performance

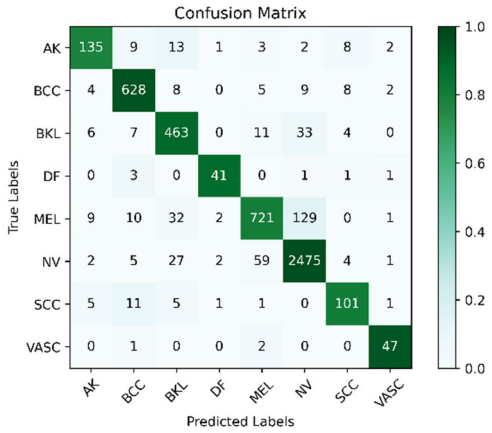


FIGURE 10. The confusion matrix of the proposed FUSCANet model on the ISIC 2019 dataset.

of the proposed model on the private dataset in the form of a confusion matrix. Table 7 presents the classification performance results of FUSCANet and the baseline models, obtained on this private dataset. Here, the proposed model holds firmly the second position after the winner (ConvNeXt Tiny).

Although FUSCANet's performance on the private dataset is slightly lower than that of ConvNeXt Tiny, the proposed model has significant advantages in terms of parameter counts. Additionally, FUSCANet outperforms all other lightweight models in classification performance, according to all five evaluation metrics. These experiments indicate that FUSCANet can accurately classify skin diseases with similar characteristics, demonstrating a good generalization ability which makes it particularly suitable for deployment on mobile devices.

In addition, we conducted experiments on a fourth dataset, namely ISIC 2019, in order to evaluate the performance of FUSCANet on this larger and more diverse dataset, and demonstrate its generalizability and scalability for use in real-world scenarios. Figure 10 presents the classification performance of the proposed model on the ISIC 2019 dataset in the form of a confusion matrix. According to the obtained results shown in Table 8, compared to the baseline models, the proposed model demonstrates the strongest classification performance according to four out five evaluation metrics used.

F. COMPARISON WITH STATE-OF-THE-ART MODELS

Next, we compared the proposed FUSCANet model with state-of-the-art models, based on their results, reported in the literature, obtained on the more representative and highly cited HAM10000 public dataset [30]. The corresponding classification performance and computational complexity comparison is shown in Table 9 (symbol "-" indicates missing data in the source).

As can be observed from Table 9, FUSCANet outperforms all state-of-the-art models across four out of five evaluation

TABLE 7. Classification performance comparison with baseline models on the private dataset.

Models	Evaluation metrics (%)				
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score
ShuffleNet V2	90.00	83.76	86.08	97.70	84.25
MobileNetV2	90.34	82.87	84.69	97.76	83.43
EfficientNet B0	90.86	83.52	87.89	97.71	85.15
MobileViT	90.69	84.04	86.75	97.68	85.18
ConvNeXt Tiny	**92.93**	**87.11**	**90.19**	**98.19**	**88.51**
FUSCANet (*PROPOSED*)	91.90	86.29	88.60	98.06	87.10

TABLE 8. Classification performance comparison with baseline models on the ISIC 2019 dataset.

Models	Evaluation metrics (%)				
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score
ShuffleNet V2	85.60	76.67	80.64	97.40	78.50
MobileNetV2	87.46	78.79	83.40	97.71	80.88
EfficientNet B0	89.98	83.30	88.07	98.17	85.48
MobileViT	90.02	85.18	87.86	98.21	86.45
ConvNeXt Tiny	90.91	84.92	**88.69**	98.32	86.66
FUSCANet (*PROPOSED*)	**91.09**	**87.36**	87.21	**98.35**	**87.20**

TABLE 9. Classification performance and computational complexity comparison with state-of-the-art models on the HAM10000 dataset.

Models	Evaluation metrics (%)					Parameter counts (M)	FLOPS (M)
	Overall accuracy	Average recall	Average precision	Average specificity	Average F1 score		
EW-FCM [35]	86.33	86.33	-	97.72	-	**1.80**	-
Fused CNNs [36]	90.85	83.81	83.75	-	83.45	-	-
G-DMN [37]	87.07	75.28	80.13	96.19	77.26	5.30	**19550.00**
EL-DLOA [38]	88.80	**89.70**	83.70	-	86.20	-	-
Custom CNN [39]	91.51					7.47	
FFN [40]	92.00	-	75.00	-	76.00	-	-
FUSCANet (*PROPOSED*)	**92.80**	87.41	**88.43**	**97.99**	**87.78**	5.27	1465.69

metrics used. The only exception is the average recall, where the winner is EL-DLOA, followed by FUSCANet. In terms of overall accuracy, FUSCANet is ahead of the second-placed model (FFN) by 0.80 percentage points. According to average precision, FUSCANet surpassed the second-best model (Fused CNNs) by 4.68 percentage points. Based on average specificity, FUSCANet outperforms the first runner-up (EW-FCM) by 0.27 percentage points. According to average F1 score, FUSCANet is ahead of the second-best model (EL-DLOA) by 1.58 percentage points.

The outstanding accuracy demonstrated by the proposed FUSCANet model means that it can correctly classify more skin lesion images, effectively reducing the misdiagnosis rate. This is crucial for clinicians, as accurate classification helps in formulating more precise treatment plans, avoiding incorrect diagnoses and unnecessary treatments. The high F1 score indicates that the proposed model strikes a good balance between precision and recall. This is particularly important for clinical screening, especially for early detection of skin lesions. The high recall value ensures that more lesions are detected in time, while high precision effectively avoids false positives, reducing the need for additional patient examinations and unnecessary anxiety. Moreover, the lower parameter counts allows the proposed model to perform skin disease classification tasks easily on mobile devices, not only enhancing the portability but also significantly reducing computational resource consumption, making it suitable for real-time diagnosis and large-scale screening. This lightweight design ensures high classification performance while meeting the stringent requirements for device load and response time in clinical applications.

G. VISUALIZATION OF EXPERIMENTAL RESULTS

For visual analysis, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) [41], an Explainable Artificial Intelligence (XAI) technique. Five skin disease images were randomly selected from the four datasets for visualization experiments, comparing the classification results of FUSCANet and the baseline (MobileViT) on which it was developed.

The visualization comparison, shown in Figure 11, highlights the attention distribution across different regions of the images, with warmer colors indicating higher attention levels. In Figure 11, the first row displays the input images, the second row shows the visualizations of MobileViT, the third row presents the visualizations after integrating the newly designed MSFA layer into the baseline (i.e., 'Baseline + MSFA' in Tables 2 and 3), the fourth row shows the visualizations after adding the newly designed ECBAM module to the baseline (i.e., 'Baseline + ECBAM' in Tables 2 and 3), and the fifth row illustrates the visualizations of the proposed FUSCANet model (i.e., 'Baseline + MSFA + ECBAM + LMV2' in Tables 2 and 3). This visual analysis experiment demonstrated the positive impact of the two attention mechanism components (MSFA and ECBAM) on the proposed model's classification results.

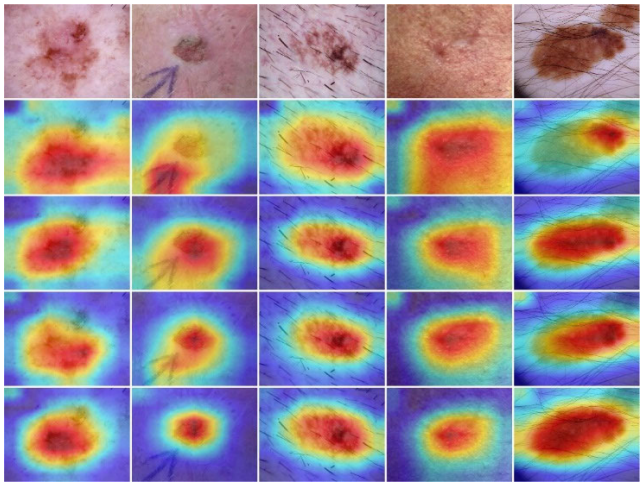


FIGURE 11. Visualization comparison of classification results.

Compared to MobileViT, the proposed FUSCANet model demonstrates a greater ability to focus attention on the lesion areas rather than on background tissues or other noisy information. Specifically, in the second column of images, MobileViT incorrectly focuses on noisy information. In other images, MobileViT fails to accurately focus on the entire lesion area; for example, in the fifth column, MobileViT only captures a portion of the lesion area. In contrast, the 'Baseline + MSFA' combination captures richer feature information, enhancing the model's understanding of the overall image structure, thereby allowing it to focus more comprehensively on the lesion areas. The 'Baseline + ECBAM' combination further refines the model's ability to precisely locate the lesion boundaries. Compared to MobileViT, the 'Baseline + MSFA + ECBAM + LMV2' combination (i.e., the proposed FUSCANet model) exhibits more accurate attention distribution, significantly concentrating on the skin lesion areas. The proposed model's ability to focus on the semantic significance of the lesions is largely attributed to the three newly designed components.

V. CONCLUSION

This paper has proposed a novel skin disease image classification model, named FUSCANet, which utilizes newly designed components, such as a Leaky MobileNetV2 (LMV2) block, an Enhanced Convolutional Block Attention Module (ECBAM), and a Multi-Scale Feature Aggregation (MSFA) layer, to extract complex feature information from skin lesion images. Experimental studies demonstrated that FUSCANet not only achieves the highest overall accuracy, compared to the baselines and state-of-the-art models, but also outperforms them according to most of the evaluation metrics used on three public datasets. Moreover, the proposed model maintains a low parameter counts, making it suitable for real-time classification in resource-constrained environments. This provides a feasible solution for deploying the proposed FUSCANet model on embedded devices and mobile applications.

In addition to its outstanding classification performance, the proposed model exhibits strong generalization ability across different data distributions. The conducted experiments showed that FUSCANet is highly generalizable and can robustly perform multi-class skin disease classification tasks in various clinical settings. Furthermore, ablation study and visualization experiments reveal that all newly designed components positively impact the classification performance of the proposed FUSCANet model, with the highest metrics achieved when all these components are integrated into the model.

However, there is still room for improvement in the lightweight design of FUSCANet. Future research will focus on further reducing the model's parameter counts to meet the higher demands of embedded devices and mobile applications. At the same time, we will ensure that the model's classification performance and generalization ability are maintained, allowing it to adapt more flexibly to different medical application scenarios.

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